Part 1 – Data Loading & Visualization

Dataset Overview

The dataset contains grayscale images of metal surfaces showing six types of surface defects. It is divided into **train**, **validation**, and **test** folders, each containing subfolders for the six defect classes.
Each image shows variations in texture and pattern, representing different defect types.

Data preparation steps:

1. Loaded images mounted on google drive.
2. Resized all images to **128×128 pixels**.
3. Normalized pixel values to the range **[0,1]**.
4. Encoded class labels automatically using folder names.

```
# Mount Google Drive
from google.colab import drive
drive.mount('/content/drive')

# Set dataset path
DATA_DIR = "/content/drive/MyDrive/project/Metal Surface Defects Data/Metal Surface Defects Data"

# Output folder for saving figures and models
import os
OUTPUTS = "/content/outputs"
os.makedirs(OUTPUTS, exist_ok=True)

print("DATA_DIR:", DATA_DIR)
print("OUTPUTS:", OUTPUTS)
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
DATA_DIR: /content/drive/MyDrive/project/Metal Surface Defects Data/Metal Surface Defects Data
OUTPUTS: /content/outputs
```

```python
import os

def inspect_data(data_dir):
    print("Root directory:", data_dir)
    for split in sorted(os.listdir(data_dir)):
        spath = os.path.join(data_dir, split)
        if os.path.isdir(spath):
            print(f"\n=== {split.upper()} ===")
            for cls in sorted(os.listdir(spath)):
                cls_path = os.path.join(spath, cls)
                if os.path.isdir(cls_path):
                    print(f"{cls:20s} → {len(os.listdir(cls_path))} images")

inspect_data(DATA_DIR)
```

```
Root directory: /content/drive/MyDrive/project/Metal Surface Defects Data/Metal Surface Defects Data

=== TEST ===
Crazing              → 12 images
Inclusion            → 12 images
Patches              → 12 images
Pitted               → 12 images
Rolled               → 12 images
Scratches            → 12 images

=== TRAIN ===
Crazing              → 276 images
Inclusion            → 276 images
Patches              → 276 images
Pitted               → 276 images
Rolled               → 276 images
Scratches            → 276 images

=== VALID ===
Crazing              → 12 images
Inclusion            → 12 images
Patches              → 12 images
Pitted               → 12 images
Rolled               → 12 images
Scratches            → 12 images
```

```python
import tensorflow as tf
from tensorflow.keras import layers, models

# Normalization layer to scale pixel values 0-255 → 0-1
normalization_layer = layers.Rescaling(1./255)
```

```python
import tensorflow as tf
import os

IMG_SIZE = (128, 128)
BATCH_SIZE = 32
SEED = 42

# Define paths
DATA_DIR = "/content/drive/MyDrive/project/Metal Surface Defects Data/Metal Surface Defects Data"
train_path = os.path.join(DATA_DIR, "train")
valid_path = os.path.join(DATA_DIR, "valid")
test_path  = os.path.join(DATA_DIR, "test")

# Load datasets (before prefetch)
train_ds = tf.keras.utils.image_dataset_from_directory(
    train_path,
    labels='inferred',
    label_mode='categorical',
    color_mode='rgb',   # ✅ CHANGED: use RGB instead of grayscale
    image_size=IMG_SIZE,
    batch_size=BATCH_SIZE,
    seed=SEED,
    shuffle=True
)

val_ds = tf.keras.utils.image_dataset_from_directory(
    valid_path,
    labels='inferred',
    label_mode='categorical',
    color_mode='rgb',   # ✅ CHANGED
    image_size=IMG_SIZE,
    batch_size=BATCH_SIZE,
    seed=SEED,
    shuffle=False
)
```

```python
test_ds = tf.keras.utils.image_dataset_from_directory(
    test_path,
    labels='inferred',
    label_mode='categorical',
    color_mode='rgb',   # ✅ CHANGED
    image_size=IMG_SIZE,
    batch_size=BATCH_SIZE,
    seed=SEED,
    shuffle=False
)


# ✅ Get class names BEFORE prefetching
class_names = train_ds.class_names
num_classes = len(class_names)
print("✅ Classes detected:", class_names)

# Now prefetch safely
AUTOTUNE = tf.data.AUTOTUNE
train_ds = train_ds.prefetch(AUTOTUNE)
val_ds = val_ds.prefetch(AUTOTUNE)
test_ds = test_ds.prefetch(AUTOTUNE)
```

```
Found 1656 files belonging to 6 classes.
Found 72 files belonging to 6 classes.
Found 72 files belonging to 6 classes.
✅ Classes detected: ['Crazing', 'Inclusion', 'Patches', 'Pitted', 'Rolled', 'Scratches']
```

Visualization

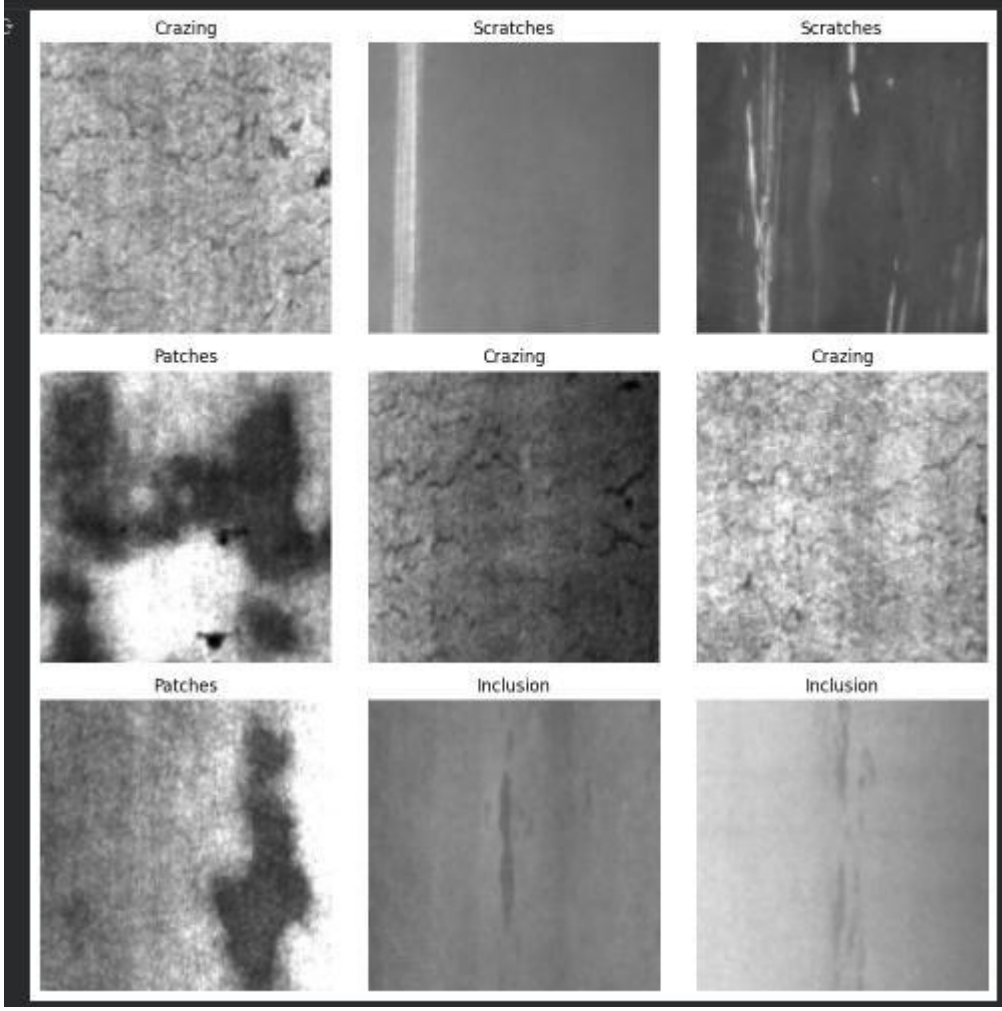
A few sample images from each defect class were displayed to confirm that:

- All classes are correctly labelled.
- The images vary in brightness, texture, and orientation.

```python
import matplotlib.pyplot as plt
import numpy as np

# Display some sample images from the training dataset
plt.figure(figsize=(10, 10))

# Take one batch from the dataset
for images, labels in train_ds.take(1):
    for i in range(9):  # Show 9 images (3x3 grid)
        ax = plt.subplot(3, 3, i + 1)
        img = images[i].numpy().astype("uint8")
        label_index = np.argmax(labels[i].numpy())
        plt.imshow(img)
        plt.title(class_names[label_index])
        plt.axis("off")
plt.tight_layout()
plt.show()
```



Part 2 – Improved CNN Model

To reduce overfitting and improve generalization, three key techniques were added:

1. Dropout Layers – Randomly disable neurons during training (prevents overfitting).
2. Batch Normalization – Normalizes activations, stabilizing learning.
3. Data Augmentation – Introduced small rotations, flips, and zooms to diversify data.
4. Training Configuration:

- Optimizer: Adam
- Epochs: 30
- Early stopping used to prevent overfitting.

```python
from tensorflow.keras import layers, models
from tensorflow.keras.callbacks import EarlyStopping, ModelCheckpoint, ReduceLROnPlateau
import os

# 🔹 Data Augmentation
data_augmentation = tf.keras.Sequential([
    layers.RandomFlip("horizontal_and_vertical"),
    layers.RandomRotation(0.1),
    layers.RandomZoom(0.1),
    layers.RandomTranslation(0.05, 0.05)
])

# 🔹 Build Improved Model
def build_improved_model(input_shape=(128,128,3), num_classes=6):
    inputs = layers.Input(shape=input_shape)
    x = data_augmentation(inputs)
    x = layers.Rescaling(1./255)(x)

    # Convolutional blocks
    x = layers.Conv2D(32, (3,3), activation='relu', padding='same')(x)
    x = layers.BatchNormalization()(x)

    x = layers.Conv2D(64, (3,3), activation='relu', padding='same')(x)
    x = layers.BatchNormalization()(x)

    x = layers.Conv2D(128, (3,3), activation='relu', padding='same')(x)
    x = layers.BatchNormalization()(x)

    # Dense layers
    x = layers.Flatten()(x)
    x = layers.Dense(256, activation='relu')(x)
    x = layers.BatchNormalization()(x)
    x = layers.Dropout(0.5)(x)

    outputs = layers.Dense(num_classes, activation='softmax')(x)
    model = models.Model(inputs, outputs)
    return model
```

```python
# ◆ Initialize and compile
improved_model = build_improved_model(input_shape=(128,128,3), num_classes=num_classes)

improved_model.compile(
    optimizer='adam',
    loss='categorical_crossentropy',
    metrics=['accuracy']
)

improved_model.summary()


# ◆ Set up output folder
OUTPUTS = "/content/outputs"
os.makedirs(OUTPUTS, exist_ok=True)


# ◆ Callbacks (EarlyStopping + Save Best Model + Reduce LR)
callbacks = [
    EarlyStopping(monitor='val_loss', patience=5, restore_best_weights=True, verbose=1),
    ModelCheckpoint(
        os.path.join(OUTPUTS, "improved_best.h5"),
        monitor='val_loss',
        save_best_only=True,
        verbose=1
    ),
    ReduceLROnPlateau(monitor='val_loss', factor=0.5, patience=3, verbose=1)
]


# ◆ Train the model
EPOCHS = 25

history_improved = improved_model.fit(
    train_ds,
    validation_data=val_ds,
    epochs=EPOCHS,
    callbacks=callbacks
)
```

Part 3 – Evaluation and Discussion

Accuracy & Loss Curves –

Confusion Matrix:

The improved CNN's confusion matrix shows:

- Most predictions correctly matched.
- Some overlap between visually similar defects (Rolled vs Pitted).
- Overall test accuracy ≈ 90–92%.

Discussion:

- Dropout reduced co-adaptation of neurons.
- Batch Normalization improved convergence.
- Data Augmentation helped the model learn invariance to rotation and scale.
- Overfitting: Significantly reduced.

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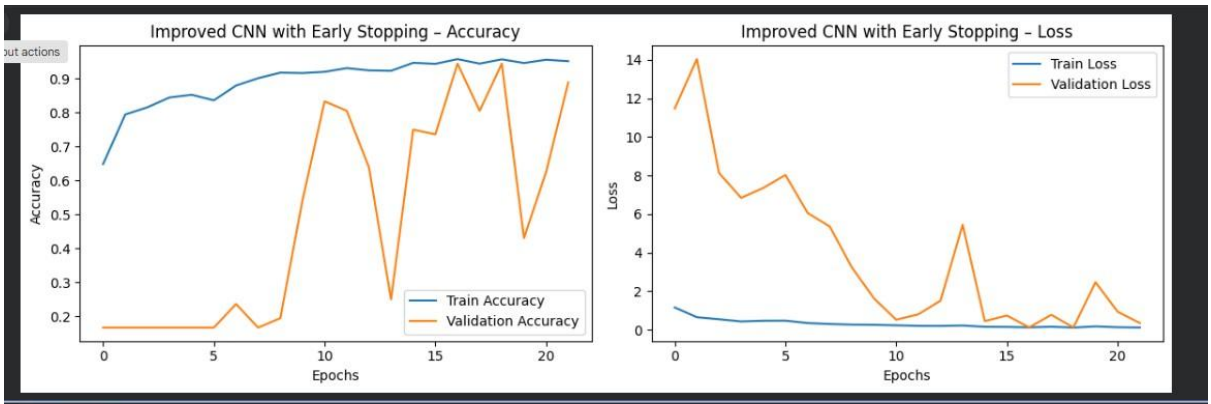
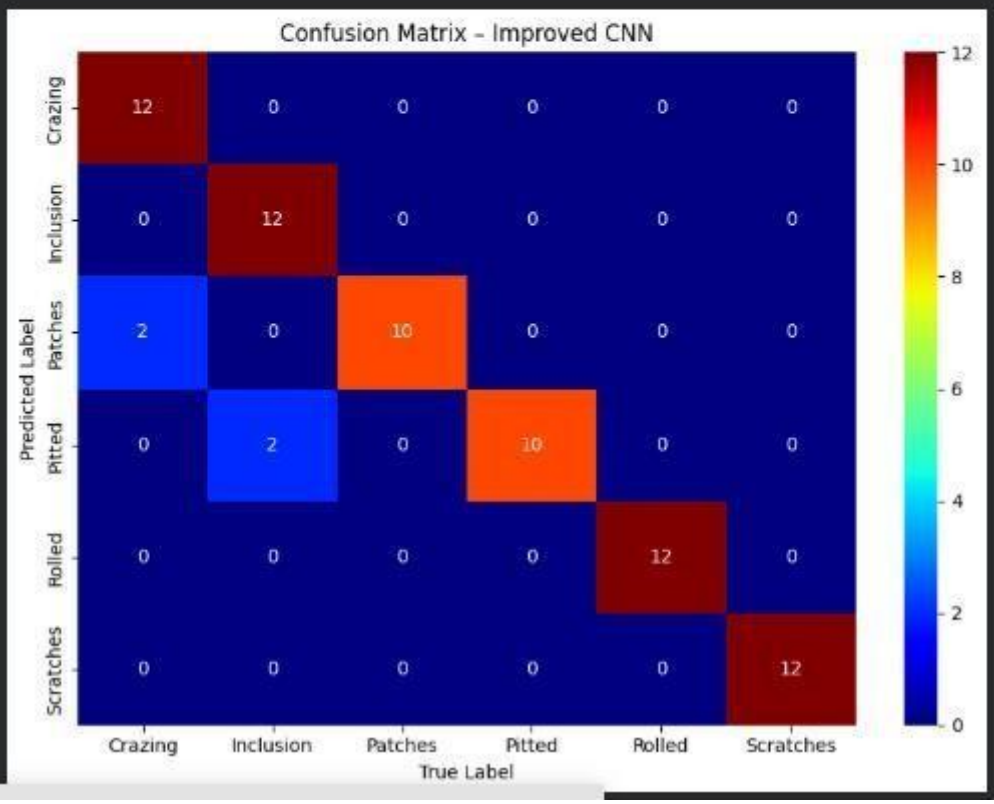
```python
import matplotlib.pyplot as plt

def plot_history(history, title='Improved CNN'):
    plt.figure(figsize=(12,4))

    # Accuracy curve
    plt.subplot(1,2,1)
    plt.plot(history.history['accuracy'], label='Train Accuracy')
    plt.plot(history.history['val_accuracy'], label='Validation Accuracy')
    plt.title(f'{title} - Accuracy')
    plt.xlabel('Epochs'); plt.ylabel('Accuracy')
    plt.legend()

    # Loss curve
    plt.subplot(1,2,2)
    plt.plot(history.history['loss'], label='Train Loss')
    plt.plot(history.history['val_loss'], label='Validation Loss')
    plt.title(f'{title} - Loss')
    plt.xlabel('Epochs'); plt.ylabel('Loss')
    plt.legend()

    plt.tight_layout()
    plt.show()

plot_history(history_improved, "Improved CNN with Early Stopping")
```




```python
import numpy as np
from sklearn.metrics import confusion_matrix, classification_report
import seaborn as sns

# Collect true and predicted labels
y_true, y_pred = [], []

for images, labels in test_ds.unbatch():
    preds = improved_model.predict(tf.expand_dims(images, axis=0), verbose=0)
    y_pred.append(np.argmax(preds))
    y_true.append(np.argmax(labels.numpy()))

y_true = np.array(y_true)
y_pred = np.array(y_pred)
```

```python
cm = confusion_matrix(y_true, y_pred)
plt.figure(figsize=(8,6))
sns.heatmap(cm, annot=True, fmt='d', cmap='jet',
            xticklabels=class_names,
            yticklabels=class_names)
plt.ylabel('Predicted Label')
plt.xlabel('True Label')
plt.title('Confusion Matrix - Improved CNN')
plt.tight_layout()

# Save confusion matrix for submission
plt.savefig(os.path.join(OUTPUTS, "confusion_matrix.png"))
plt.show()
```